

P29 B(11)

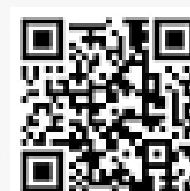
$$x^2 - x - 2 \neq 0$$

$$x + 2 \neq 0 \Rightarrow x \neq -2$$

$$\rightarrow (x - 2)(x + 1) \neq 0$$

$$x - 2 \neq 0, x + 1 \neq 0 \Rightarrow x \neq 2, -1$$

$$\frac{x^2 - 4}{x^2 - x - 2} \cdot \frac{x^2 - 1}{x + 2} = \frac{(x + 2)(x - 2)}{(x - 2)(x + 1)} \cdot \frac{(x + 1)(x - 1)}{x + 2} = x - 1$$



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$$x^3 + \frac{1}{x^3} = ?$$

$$\textcircled{1} \times \textcircled{2} \quad (x + \frac{1}{x}) (x^2 + \frac{1}{x^2}) = 3 \times 7$$

$$\textcircled{x^3} + \underbrace{x + \frac{1}{x}} + \textcircled{\frac{1}{x^3}} = 21$$

$$(x^3 + \frac{1}{x^3}) + 3 = 21$$

$$x^3 + \frac{1}{x^3} = 21 - 3 = 18$$

P30 C (2)

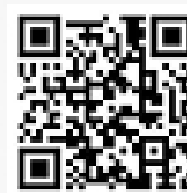
$$x + \frac{1}{x} = 3 \quad \textcircled{1}$$

$$x^2 + \frac{1}{x^2} = ? \quad \textcircled{2}$$

$$\textcircled{1}^2 : (x + \frac{1}{x})^2 = 9$$

$$\textcircled{x^2} + 2 \times 1 \times \textcircled{\frac{1}{x^2}} = 9$$

$$x^2 + \frac{1}{x^2} = 9 - 2 = 7$$



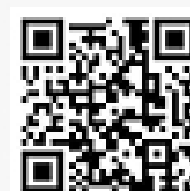
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P30 C (4)

$$= 3 \times 7$$
$$(a+b)^2 = a^2 + 2ab + b^2$$
$$a^2 - b^2 = (a+b)(a-b)$$
$$= 21$$

$$x^4 + 4x^2 + 16$$
$$= (x^2)^2 + 2 \cdot 4 \cdot x^2 + 4^2 - 4x^2$$
$$= (x^2 + 4)^2 - (2x)^2$$
$$= (x^2 + 4 - 2x)(x^2 + 4 + 2x)$$



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P30 C(5)

正数:  $>0$  负数:  $<0$

$$\begin{aligned} & x^2 + 2x + 2 \\ &= \frac{x^2 + 2 \cdot x \cdot 1 + 1^2 + 1}{1} \\ &= (x+1)^2 + 1 \\ &\geq 0 + 1 \\ &= 1 \end{aligned}$$

$$\square^2 \geq 0$$

$$\square = 0 \quad \square^2 = 0$$

$$\square > 0 \quad \square^2 > 0$$

$$\square < 0 \quad \square^2 > 0$$

负负得正



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P30 C (5)

正数:  $>0$  负数:  $<0$

$$\begin{aligned} & x^2 + 2x + 2 \\ &= \frac{x^2 + 2 \cdot x \cdot 1 + 1^2 + 1}{(x+1)^2 + 1} \\ &\geq 0 + 1 \\ &= 1 \end{aligned}$$

平方差

P48 2.4.1 (1)

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

$$(x=0) \text{ 或 } x^2 - 4 = 0$$

$$(x+2)(x-2) = 0$$

$$x+2=0 \text{ 或 } x-2=0$$

$$(x=-2 \text{ 或 } 2)$$



P48 2.4.1(2)

$$x^3 + 2x^2 - x - 2 = 0$$

$$(x^3 + 2x^2) - (x + 2) = 0$$

$$\underbrace{x^2(x+2)} - \underbrace{(x+2)} = 0$$

$$(x+2)(x^2 - 1) = 0$$

平方差

$$x+2=0 \text{ 或 } x^2-1=0$$

$$x = -2$$

$$x = 1 \text{ 或 } -1$$

P48 2.4.1(1)

$$x^3 - 4x = 0$$

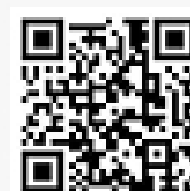
$$x(x^2 - 4) = 0$$

$$x = 0 \text{ 或 } x^2 - 4 = 0$$

$$(x+2)(x-2) = 0$$

$$x+2=0 \text{ 或 } x-2=0$$

$$x = -2 \text{ 或 } 2$$



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$$\underline{x^2=1 \text{ 或 } x^2=9}$$

$$\underline{x=1 \text{ 或 } -1}$$

$$\underline{x=3 \text{ 或 } -3}$$

P48 2.4.1 (3)

$$(x^2)^2 - 10x^2 + 9 = 0$$

把  $x^2$  当作整体

$$t = x^2$$

$$t^2 - 10t + 9 = 0$$

$$(t-1)(t-9) = 0$$

$$t=1 \text{ 或 } t=9$$



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$$(x-1)^2=1 \text{ 或 } (x-1)^2=4$$
$$x-1=\pm 1 \quad x-1=\pm 2$$
$$x=2 \text{ 或 } 0 \quad x=3 \text{ 或 } -1$$

P48 2.4.1 (4)

$$(x-1)^4 - 5(x-1)^2 + 4 = 0$$

令  $t = (x-1)^2$  换元法

$$t^2 - 5t + 4 = 0$$
$$(t-1)(t-4) = 0$$
$$t=1 \text{ 或 } t=4$$



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P58 A(6)

$$\frac{3x^2}{a} - \frac{6x}{b} + \frac{4}{c} = 0$$

$$\Delta = b^2 - 4ac$$

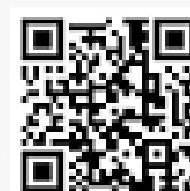
$$= (-6)^2 - 4 \times 3 \times 4$$

$$= 36 - 48 = -12$$

$\Delta < 0$ : 没有实数根

$\Delta = 0$ : 有1个

$\Delta > 0$ : 有2个



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P59 B (2)

$$x_1, x_2: \underset{a}{1}x^2 - \underset{b}{4}x - \underset{c}{2} = 0$$

韦达定理  $x_1 + x_2 = -\frac{b}{a} = 4$

$$x_1 x_2 = \frac{c}{a} = -2$$

$$x_1^2 x_2 + x_1 x_2^2$$

$$= x_1 x_2 (x_1 + x_2) = -8$$

P59 B (3)

$$1x^2 - kx + k = 0$$

$$a=1, b=-k, c=k$$

$$\Delta = b^2 - 4ac$$

$$= (-k)^2 - 4 \times 1 \times k = 0$$

$$k^2 - 4k = 0$$

$$k=0 \text{ 或 } 4$$

$$\Delta = 0: \text{有1个}$$

$$\Delta > 0: \text{有2个}$$



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